KERAS TRAINING PARAMETERS

Simple definitions, examples, calculations and a complete practical example

1. What is Dense in Keras?

Definition: A Dense layer is a fully connected neural network layer. Each neuron connects to all outputs from the previous layer.

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

model = Sequential([
    Input(shape=(3,)),
    Dense(8, activation="relu"),
    Dense(1, activation="sigmoid")
])

model.summary()
```

Explanation: Input(shape=(3,)) means three input features. Dense(8) creates a hidden layer with eight neurons. Dense(1) creates one output neuron. Sigmoid is commonly used for binary classification.

Remember: Dense(8) means eight neurons, not eight layers. The predicted class depends on the target labels used in the dataset.

2. What is Batch Size?

Definition: Batch size is the number of training samples processed before one optimizer update.

```
model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=100
)
```

Example: If there are 1,000 training samples and batch_size=100, the model processes 100 samples per batch.

Batches per epoch = Training samples / Batch Size
1,000 / 100 = 10 batches per epoch.

Dataset size	Batch size	Batches per epoch
1,000	10	100
1,000	50	20
1,000	100	10
1,000	200	5

Small batch: More optimizer updates per epoch and often noisier gradients. **Large batch:** Fewer updates per epoch and usually more memory required. A common starting point is 32.

3. What is Validation Split?

Definition: validation_split specifies the fraction of the data passed to model.fit() that Keras reserves for validation. Validation data monitors performance and is not used to update model weights.

```
model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32,
    validation_split=0.2
)
```

Calculation: If X_train contains 800 samples and validation_split=0.2, then 800 × 0.20 = 160 validation samples. The remaining 640 samples are used for gradient updates. A separately held-out test dataset remains unchanged.

Part	Number of samples
Data passed to fit()	800
Training subset	640
Validation subset	160
Separate test set	Unchanged

Important: With NumPy arrays or tensors, Keras normally takes the last fraction of the supplied arrays for validation before shuffling the remaining training data. Split and prepare data carefully to avoid data leakage.

4. What is Verbose?

Definition: verbose controls how much progress information Keras displays while training. It does not change the model's learning algorithm.

Value	Meaning	Typical display
verbose=0	Silent	No normal progress output
verbose=1	Progress bar	Progress bar and metrics
verbose=2	Compact	Usually one line per epoch

```
model.fit(X_train, y_train, epochs=5, verbose=0)
model.fit(X_train, y_train, epochs=5, verbose=1)
model.fit(X_train, y_train, epochs=5, verbose=2)
```

5. All Four Concepts in One Example

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

model = Sequential([
    Input(shape=(3,)),
    Dense(8, activation="relu"),
    Dense(1, activation="sigmoid")
])

model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32,
    validation_split=0.2,
    verbose=1
)
```

Parameter summary: Dense(8) = eight neurons in the hidden layer; epochs=10 = up to ten passes through the training subset; batch_size=32 = up to 32 samples per batch; validation_split=0.2 = reserve 20% of the supplied training data for validation; verbose=1 = display the progress bar.

Quick Revision

Dense: fully connected layer.
Batch size: samples processed before an optimizer update.
Validation split: fraction of the supplied training data reserved for validation.
Verbose: controls how much training progress is displayed.